

the data to learn and make inferences, gets smarter over time and need not be trained in the traditional way. It takes a novel approach to computing via asynchronous spiking.

The brain's neural networks relay information with pulses or spikes, modulate the synaptic strengths or weight of the interconnections based on timing of these spikes, and store these changes locally at the interconnections. Intelligent behaviours emerge from the cooperative and competitive interactions between multiple regions within the brain's neural networks and its environment.

The Loihi research test chip includes digital circuits that mimic the brain's basic mechanics, making machine learning faster and more efficient while requiring lower compute power.

The potential benefits from self-learning chips are limitless. One example provides a person's heartbeat reading under various conditions—after jogging, following a meal or before going to bed—to a neuromorphic-based system that parses the data to determine a normal heartbeat. The system can then continuously monitor incoming heart data in order to flag patterns that do not match the normal pattern. The system could be personalised for any user.

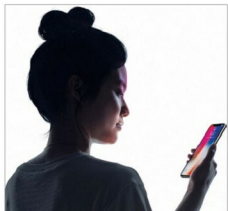
This type of logic could also be applied to other use cases, like cybersecurity where an abnormality or difference in data streams could identify a breach or a hack since the system has learned the normal behaviour under various contexts.

## FaceID cracked with a \$150 mask

Days after the release of iPhone X, a cybersecurity firm has been able to bypass its security measure of FaceID just by using a \$150 face mask. A video posted by the firm showed that they were able to hack FaceID by using a composite mask of 3-D-printed plastic, silicone, makeup and simple paper cutouts, which in combination tricked the iPhone X into unlocking. The experiment showed that in order to

bypass FaceID even half of the face was enough. FaceID recognised half the face and allowed the user to unlock the phone.

In the video, a staff member pulls off a cloth from a face mask which



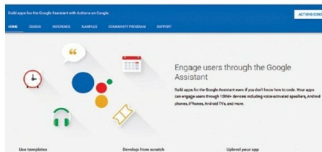
FaceID on iPhone X authenticates users with facial recognition

is placed in front of the iPhone 10. The phone instantly unlocks when the mask is placed in front of it. The mask had sculpted silicone nose, two-dimensional eyes and lips printed on paper, which was mounted on a 3D printed plastic frame.

Ankush Johar, Infosec Ventures director and partner, said, "With FaceID, your sole authentication medium is always out in the open, up for grabs. If your phone can use an infrared scan to trace every inch of your face, then so can any other hardware and this event simply affirms the theory making your data a few bucks away for any malicious hacker." He suggests FaceID users "to simply use the conventional passcodes as they are their safest bet in the absence of TouchID. Apple, on the other hand, while fixing this issue, can parallelly host a bug hunting program for the FaceID particularly."

## Developers invited to build apps for Google Assistant

Google is inviting developers and companies to build voice-enabled apps to engage with Indian users through Actions on Google, the developer platform for Google Assistant. For anyone who wants to build for the Assistant, resources such as developer tools, documentation and a simulator are available on Actions on Google developer website, making it easy to create, test and deploy voice-enabled solutions. Eligible developers can also join the Google Assistant Developer Community Program to get started with building apps for the Google Assistant.



Actions on Google platform

With the ability to create apps on the Google Assistant, users in India will soon have easy and fast access to all types of interactions, using nothing more than their voice. When apps are built to work with the Assistant, users can simply tell Assistant to connect with the app using a voice command—whether it's on supported Android phones or iPhones. Also this doesn't need anything extra to be installed—users can interact with their favourite app by saying "Ok Google, talk to..." to interact with that app. This platform will give more Indians the help they need using the convenience of a voice command at home or on-the-go—from the morning rush hour to the weekend unwind.